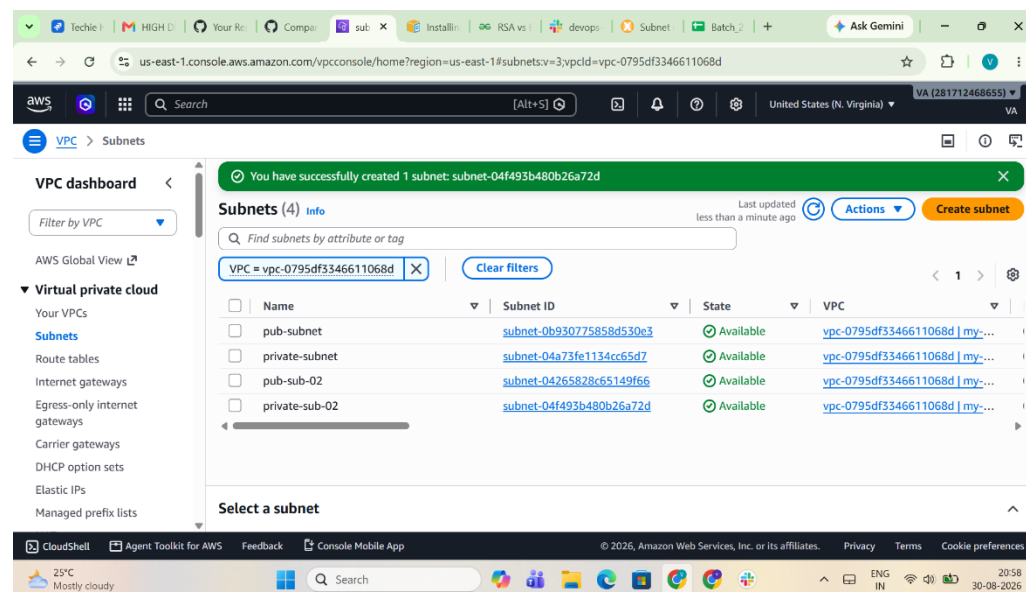


Task on VPC

1. Create VPC with 2 private and 2 public subnets.

Implementation:

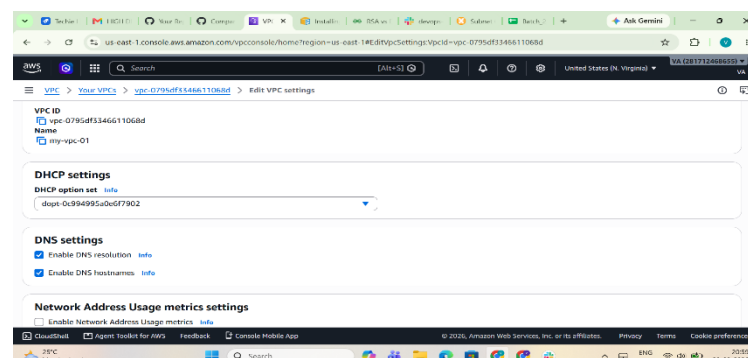
1. Created a VPC with a suitable CIDR block.
2. Created 2 public subnets in different Availability Zones.
3. Created 2 private subnets in different Availability Zones.
4. Verified the subnet CIDR ranges and VPC associations.



2. Enable DNS Hostname in VPC.

Implementation:

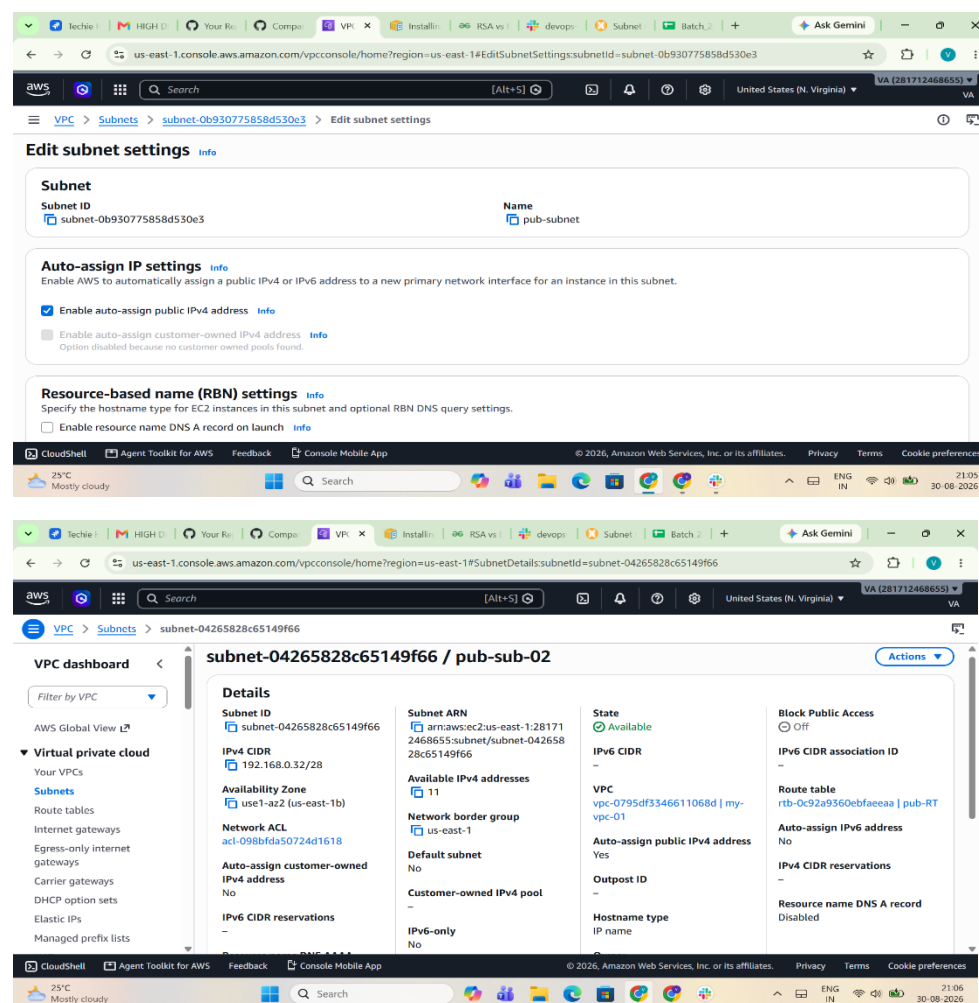
1. Opened the VPC settings.
2. Enabled DNS Resolution for the VPC.
3. Enabled DNS Hostnames for the VPC.
4. Verified that the DNS settings were enabled successfully.



3. Enable Auto Assign Public IP in 2 public subnets.

Implementation:

1. Opened the subnet configuration.
2. Selected the first public subnet.
3. Enabled Auto-Assign Public IPv4 Address.
4. Repeated the configuration for the second public subnet.
5. Verified that both public subnets were configured to assign public IP addresses automatically.

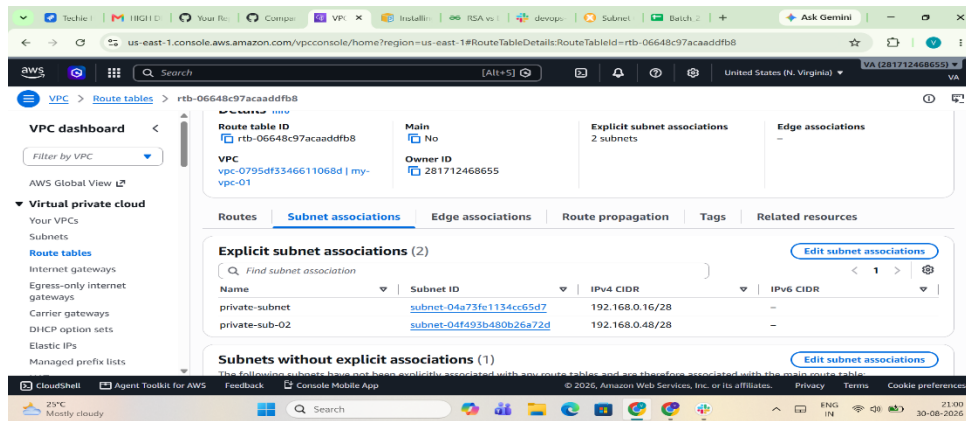


4. Add 2 private subnets in private route table.

Implementation:

1. Created a private route table for the VPC.
2. Associated the first private subnet with the private route table.
3. Associated the second private subnet with the private route table.

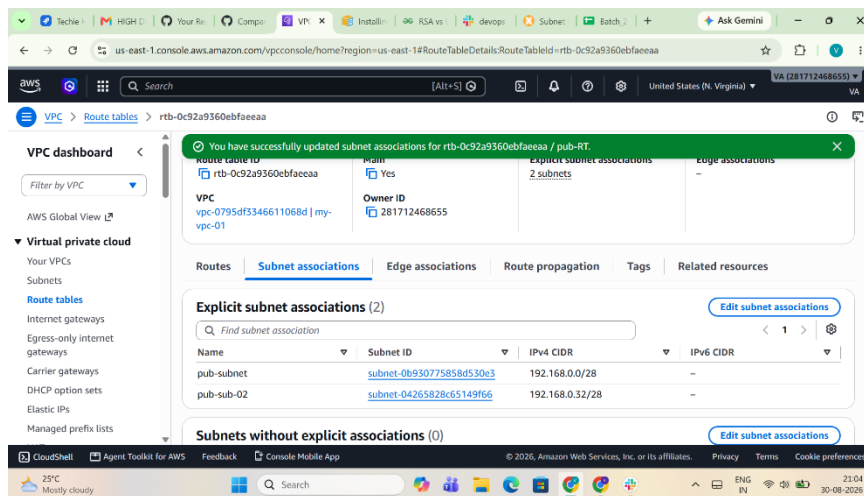
4. Verified both private subnets were associated successfully.



5. Add 2 public subnets in public route table.

Implementation:

1. Created a public route table for the VPC.
2. Associated the first public subnet with the public route table.
3. Associated the second public subnet with the public route table.
4. Verified both public subnets were associated successfully.



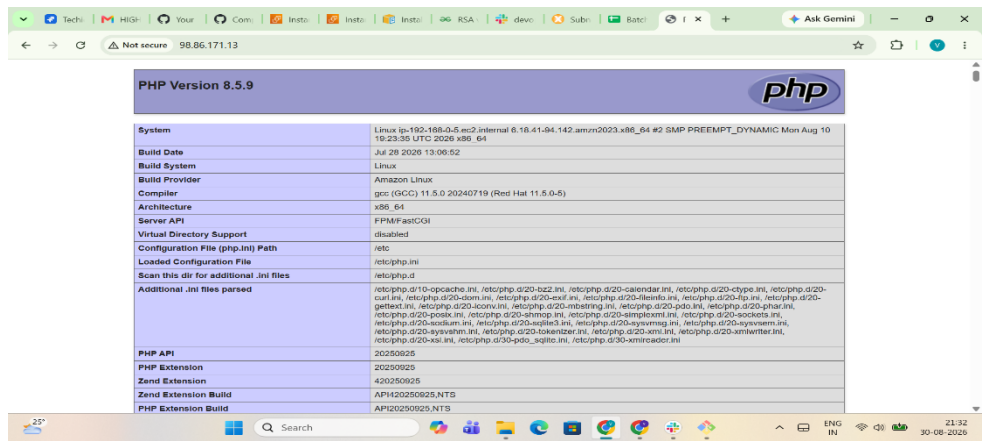
6. Public route table will have the routes to internet and local.

Implementation:

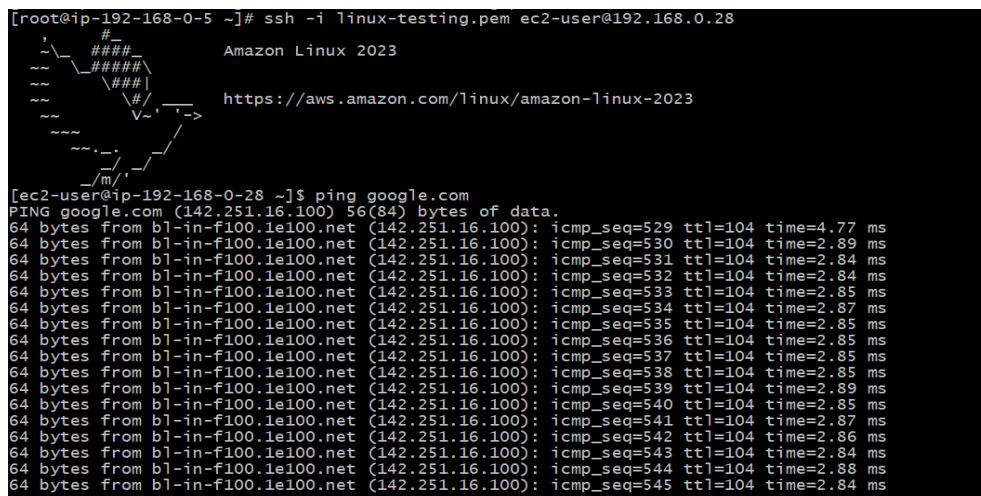
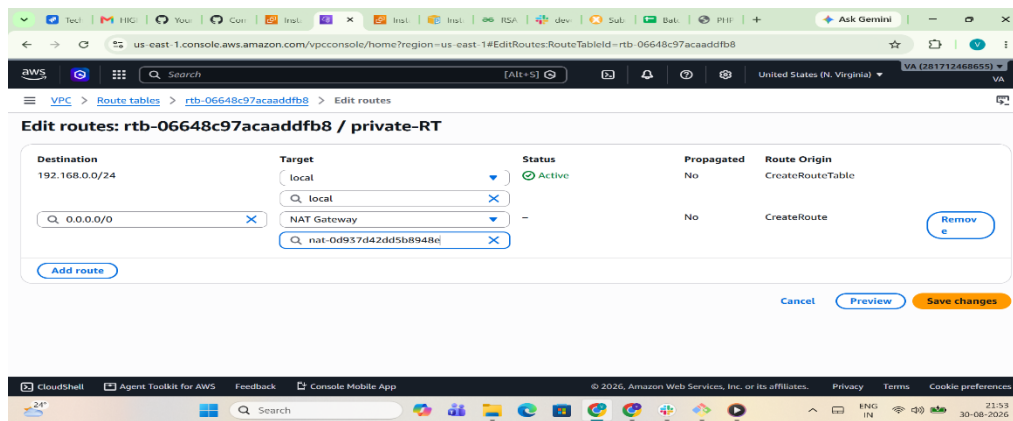
1. Opened the public route table.
2. Verified the default local route for VPC communication.
3. Added a 0.0.0.0/0 route pointing to the Internet Gateway.
4. Verified that the public route table provided internet connectivity.


```
[root@ip-192-168-0-5 ~]# php -v
PHP 8.5.9 (cli) (built: Jul 28 2026 13:06:52) (NTS gcc x86_64)
Copyright (c) The PHP Group
Built by Amazon Linux
Zend Engine v4.5.9, Copyright (c) Zend Technologies
    with Zend OPcache v8.5.9, Copyright (c), by Zend Technologies
```

```
[root@ip-192-168-0-5 ~]# with Zend OPcache v8.5.9, Copyright (c), by Zend Technologies
[root@ip-192-168-0-5 ~]# sudo dnf install httpd php -y
[sudo systemctl] enable --now httpd
Last metadata expiration check: 0:01:19 ago on Sun Aug 30 15:56:18 2026.
Package httpd-2.4.68-1.amzn2023.0.1.x86_64 is already installed.
Package php8.5-8.5.9-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-192-168-0-5 ~]# sudo vi /var/www/html/index.php
[root@ip-192-168-0-5 ~]#
```



1. Allocated an Elastic IP address for the NAT Gateway.
2. Created a NAT Gateway in the public subnet.
3. Updated the private route table with a 0.0.0.0/0 route through the NAT Gateway.
4. Launched/configured the private EC2 instance without a public IP.
5. Configured the security groups to allow required communication between the public and private instances.
6. Tested outbound internet connectivity from the private EC2 through the NAT Gateway.



1. Connected to the private EC2 instance through the public EC2/bastion host.
2. Installed Java required for Tomcat.
3. Downloaded and extracted Apache Tomcat.
4. Configured the Tomcat installation and permissions.
5. Started the Tomcat server on port 8080.

- Downloaded the sample WAR application.
- Copied the WAR file into the Tomcat webapps directory.
- Verified that Tomcat automatically deployed the sample application.

```
[root@ip-192-168-0-28 webapps]# wget https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war
--2026-08-30 16:37:19-- https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war
Resolving tomcat.apache.org (tomcat.apache.org)... 151.101.2.132, 20414642:644
Connecting to tomcat.apache.org (tomcat.apache.org)[151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4606 (4.5K)
Saving to: 'sample.war'

sample.war
100%[=====] 4.50K --.-KB/s in 0s

2026-08-30 16:37:19 (52.6 MB/s) - 'sample.war' saved [4606/4606]

[root@ip-192-168-0-28 webapps]# ls -lh sample.war
-rw-r--r-- 1 root root 4.5K Mar 31 2012 sample.war
[root@ip-192-168-0-28 webapps]# chown tomcat:tomcat sample.war
[root@ip-192-168-0-28 webapps]# su -s /bin/bash tomcat -c "/opt/tomcat/apache-tomcat-9.0.121/bin/shutdown.sh"
Using CATALINA_BASE: /opt/tomcat/apache-tomcat-9.0.121
Using CATALINA_HOME: /opt/tomcat/apache-tomcat-9.0.121
Using CATALINA_TMPDIR: /opt/tomcat/apache-tomcat-9.0.121/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/tomcat/apache-tomcat-9.0.121/bin/bootstrap.jar:/opt/tomcat/apache-tomcat-9.0.121/bin/tomcat-juli.jar
Using CATALINA_OPTS:
NOTE: Picked up JDK_JAVA_OPTIONS:  --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.lang.invoke=ALL-UNNAMED --add-opens=java.base/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java.base/java.util.concurrent=ALL-UNNAMED --add-opens=java.rmi.transport=ALL-UNNAMED
[root@ip-192-168-0-28 webapps]# su -s /bin/bash tomcat -c "/opt/tomcat/apache-tomcat-9.0.121/bin/startup.sh"
Using CATALINA_BASE: /opt/tomcat/apache-tomcat-9.0.121
Using CATALINA_HOME: /opt/tomcat/apache-tomcat-9.0.121
Using CATALINA_TMPDIR: /opt/tomcat/apache-tomcat-9.0.121/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/tomcat/apache-tomcat-9.0.121/bin/bootstrap.jar:/opt/tomcat/apache-tomcat-9.0.121/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-192-168-0-28 webapps]# ls -la
total 56
drwxr-xr-x 9 tomcat tomcat 113 Aug 30 16:37 .
drwxr-xr-x 9 tomcat tomcat 16384 Aug 30 16:27 ..
drwxr-xr-x 3 tomcat tomcat 16384 Aug 30 16:27 ROOT
drwxr-xr-x 16 tomcat tomcat 16384 Aug 30 16:27 docs
drwxr-xr-x 7 tomcat tomcat 99 Aug 30 16:27 examples
drwxr-xr-x 6 tomcat tomcat 79 Aug 30 16:27 host-manager
drwxr-xr-x 6 tomcat tomcat 114 Aug 30 16:27 manager
drwxr-xr-x 5 tomcat tomcat 86 Aug 30 16:37 sample
-rw-r--r-- 1 tomcat tomcat 4606 Mar 31 2012 sample.war
[root@ip-192-168-0-28 webapps]#
```

```
[root@ip-192-168-0-28 webapps]# curl -I http://192.168.0.28:8080
HTTP/1.1 200
Content-Type: text/html; charset=UTF-8
Date: Sun, 30 Aug 2026 16:43:48 GMT

[root@ip-192-168-0-28 webapps]# curl http://192.168.0.28:8080/sample/
<html>
<head>
<title>Sample "Hello, World" Application</title>
</head>
<body bgcolor=white>

<table border="0">
<tr>
<td>

</td>
<td>
<h1>Sample "Hello, World" Application</h1>
<p>This is the home page for a sample application used to illustrate the
source directory organization of a web application utilizing the principles
outlined in the Application Developer's Guide.
</p>
</td>
</tr>
</table>

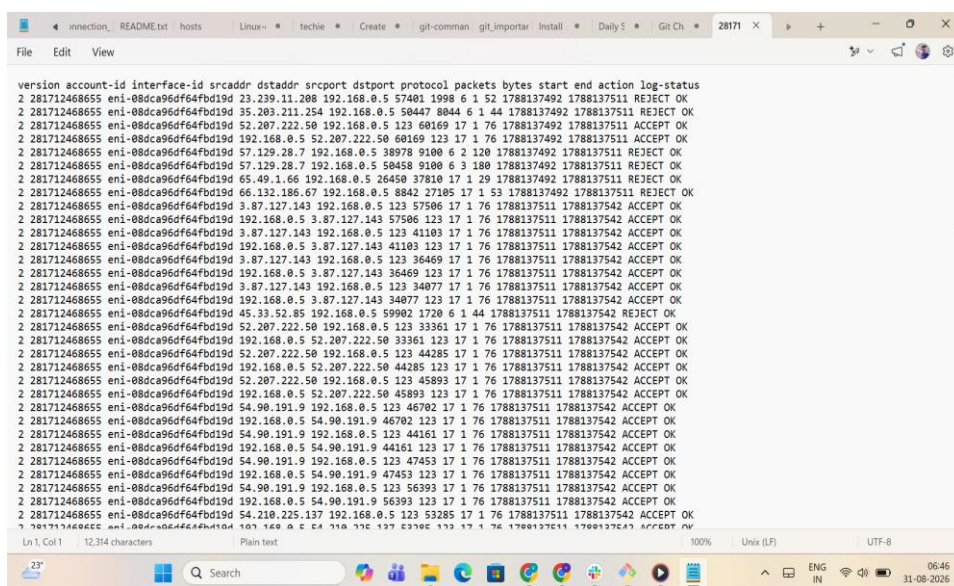
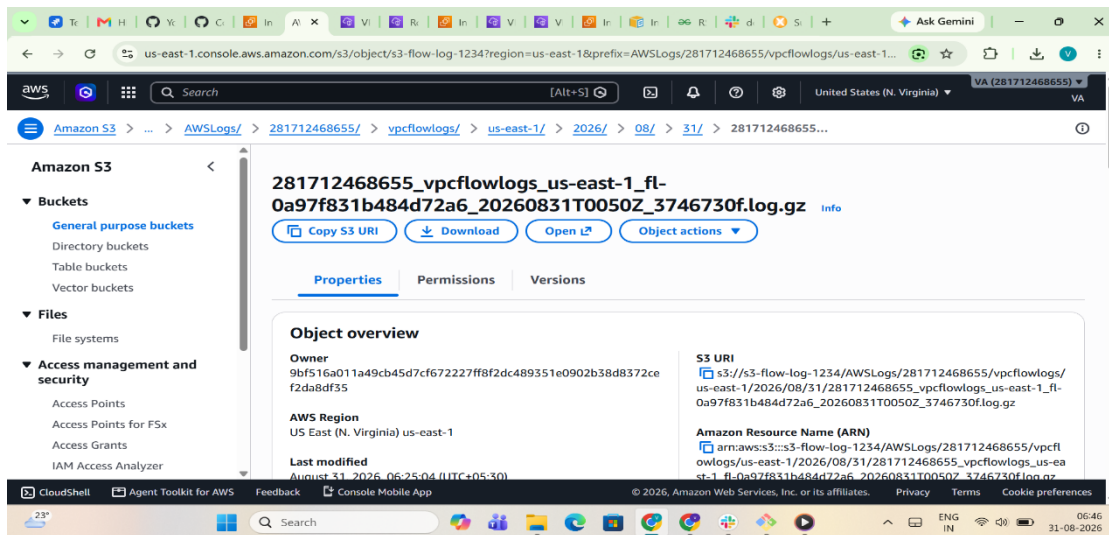
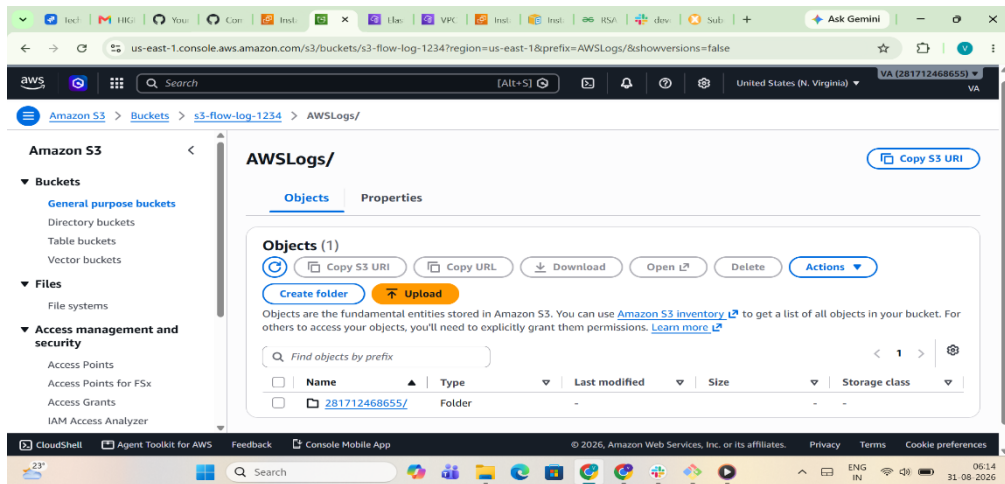
<p>To prove that they work, you can execute either of the following links:
<ul>
<li>To a <a href="hello.jsp">JSP page</a>.
<li>To a <a href="hello">servlet</a>.
</li>
</ul>
</p>
</body>
</html>
[root@ip-192-168-0-28 webapps]#
```

10. Configure VPC flow logs and store the logs in S3 and CloudWatch.

Implementation:

- Enabled VPC Flow Logs for the required VPC.
- Configured CloudWatch Logs as a flow-log destination.
- Created/configured the required IAM permissions for CloudWatch log delivery.
- Configured Amazon S3 as a storage destination for flow logs.
- Generated network traffic between the EC2 instances.

6. Verified the captured network traffic logs in CloudWatch and S3.



Cloudwatch

CloudWatch > Log management

Log group "vpc-flowlog-cloudwatch" has been created.

Log groups (1)

Log group	Log class	Anomaly d...	Del...	Bea...	D...
vpc-flowlog-cloudwatch	Standard	Configure	Off	Off	-

Identity and Access Management (IAM)

Policies (1571)

Policy vpc-flow-log-cloudwatch created.

Policy name	Type	Used as
vpc-flow-log-cloudwatch	Customer managed	None

Identity and Access Management (IAM)

Roles (5)

Role vpcflowlog-role created.

Role name	Trusted entities	Last activity
vpcflowlog-role	AWS Service: vpc-flow-logs	-

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#FlowLogDetails:flowLogId=fl-013bbd98d8fb73e80

Search [Alt+S]

United States (N. Virginia) VA (281712468655)

VPC > Your VPCs > fl-013bbd98d8fb73e80

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Successfully created flow log for the following resource:

vpc-0795df3346611068d

fl-013bbd98d8fb73e80 / cloudwatch-flowlog

Details

Flow Log ID fl-013bbd98d8fb73e80	Destination Type cloud-watch-logs	Traffic Type All	File Format -
Name cloudwatch-flowlog	Destination Name vpc-flowlog-cloudwatch	Max Aggregation Interval 1 minute	Hive Compatible Partitions -
State Active	IAM Role arn:aws:iam::281712468655:role/vpc-flowlog-role	Log Format Default	Partition Logs -
Creation Time Monday, August 31, 2026 at 13:12:51 GMT+5:30	Cross Account IAM Role -	Tag field specifications -	

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#VpcDetails:VpcId=vpc-0795df3346611068d

Search [Alt+S]

United States (N. Virginia) VA (281712468655)

VPC > Your VPCs > vpc-0795df3346611068d

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

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Carrier gateways

DHCP option sets

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Managed prefix lists

Main network ACL
acl-098bfa50724d1618

Default VPC
No

IPv4 CIDR
192.168.0.0/24

IPv6 CIDR (Network border group)
-

Network Address Usage metrics
Disabled

Route 53 Resolver DNS Firewall rule groups
-

Encryption control ID
-

Encryption control mode
-

RT
IPv6 pool
-

Owner ID
281712468655

Resource map CIDRs **Flow logs** Tags Related resources Integrations

Flow logs (1)

Search

Name	Flow log ID	Traffic type	De
cloudwatch-flowlog	fl-013bbd98d8fb73e80	All	clk

us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#logsV2-log-groups/log-group/vpc-flowlog-cloudwatch/log-events/eni-08dca96df64fbd19d-all

Search [Alt+S]

United States (N. Virginia) VA (281712468655)

CloudWatch > Log management > vpc-flowlog-cloudwatch > eni-08dca96df64fbd19d-all

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Container Insights

Database Insights

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Display

1m 1h UTC timezone

Timestamp	Message
2026-08-31T07:49:52.000Z	2 281712468655 eni-08dca96df64fbd19d 85.217.140.44 192.168.0.5 39407 13843 6 1 52 1788162592...
2026-08-31T07:49:52.000Z	2 281712468655 eni-08dca96df64fbd19d 192.168.0.5 85.217.140.44 13843 39407 6 1 40 1788162592...
2026-08-31T07:49:52.000Z	2 281712468655 eni-08dca96df64fbd19d 130.12.180.246 192.168.0.5 37006 8746 6 1 40 1788162592...
2026-08-31T07:49:52.000Z	2 281712468655 eni-08dca96df64fbd19d 192.168.0.5 130.12.180.246 8746 37006 6 1 40 1788162592...
2026-08-31T07:49:52.000Z	2 281712468655 eni-08dca96df64fbd19d 162.216.149.139 192.168.0.5 52020 22001 6 1 44 1788162592...
2026-08-31T07:49:52.000Z	2 281712468655 eni-08dca96df64fbd19d 192.168.0.5 162.216.149.139 22001 52020 6 1 40 1788162592...
2026-08-31T07:50:22.000Z	2 281712468655 eni-08dca96df64fbd19d 192.168.0.5 45.135.193.115 0 1 1 95 1788162592...

Back to top